

FRANK LIU

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SKILLS

Programming Languages: JavaScript, Typescript, Java, Python, Bash, SQL, Go, HTML/CSS

Framework: Angular, React, Vue, Express, Node.js/NPM, Spring, Flask, jQuery

Tools: NoSQL, Git, Docker, Jira, Jenkins, Redis, Linux OS, Tailwind

PROFESSIONAL EXPERIENCE

Remote Technology Solutions – Remote

09/2024 – Present

Lead Full Stack Engineer

- Boosted real-time terminal performance 9× by redesigning canvas rendering and throttling reflows to sustain 30+ concurrent WebSocket Telnet sessions without frame drops.
- Eliminated Vue/Pinia async race conditions by refactoring store patterns, improving real-time UI stability and reducing error reports by 80%.
- Directed architecture for a platform serving 10,000+ users across 30 countries by leading design reviews and code standards to align delivery with roadmap milestones.

Level Access – Ottawa, ON

05/2022 – 09/2024

Full Stack Engineer

- Partnered with auditors and PMs in an Agile SaaS startup environment to deliver WCAG 2.2 compliance across core workflows, resolving 20+ accessibility blockers and increasing accessible coverage to 95%.
- Scaled enterprise web crawler capacity 10× (1K → 10K pages per job) by redesigning architecture with RabbitMQ job orchestration and Redis caching.
- Cut MongoDB query latency from 30s to <1s on 700M+ records by profiling with Datadog/AWS logs and rewriting queries/indexes, lowering p95 API response times by 88%.
- Built responsive, mobile-friendly Angular components with RxJS/reactive forms based on Figma mockups and implemented CI for tests (Jasmine/Karma, Cypress), increasing unit coverage from 76% to 82%.
- Shipped secure Express microservices with authn/authz and RBAC, enabling third-party integrations via public REST APIs while maintaining PII auditability.

Red Canari – Ottawa, ON

05/2021 - 08/2021

Full Stack Engineer

- Architected a proxy-layer database in a React front-end, applying the SQL hook design pattern and efficient data structures with live schema enforcement, uncovering 40+ critical data integrity failures.

Ericsson – Ottawa, ON

05/2020 - 04/2021

Software Engineer

- Automated validation of Kubernetes JSON schemas by developing a Python/Kafka pipeline integrated into CI/CD, uncovering 5,000+ inconsistencies.

Carleton University – Ottawa

05/2018 - 08/2018

Software Engineer

- Applied machine learning expertise in Java, working with DL4J and Keras to enable cross-framework AI development.

EDUCATION

Bachelor of Engineering: Software Engineer, Co-op

09/2017 - 04/2022

Carleton University – Ottawa, Ontario 3.8 GPA